

Underwater Noise Impact Assessment

Sound source: Sercel Micro-GI Air Gun
Animal group: VHF

1 General information

1.1 Animal group VHF

Very High-frequency (VHF) Cetaceans including Delphinidae (Cephalorhynchus spp.; Lagenorhynchus cruciger, L. australis); Phocoenidae (Neophocaena spp., Phocoena spp., Phocoenoides); Iniidae (Inia); Kogiidae (Kogia); Lipotidae (Lipotes); Pontoporiidae (Pontoporia)

Permanent and Temporary Threshold Shift (PTS, TTS) limits are determined with the more conservative measure of the dual exposure metrics of unweighted zero-peak Sound Pressure Level as compiled from Southall et al. (2007, 2019); BOEM (2014a,b); Statoil ASA (2015); Tougaard et al. (2014, 2015, 2016); Schack et al. (2019):

$$SPL_{uw,0-peak}(PTS) = 202 \text{ dB re } 1 \mu Pa$$

$$SPL_{uw,0-peak}(TTS) = 196 \text{ dB re } 1 \mu Pa$$

and the weighted Sound Exposure Level:

$$SEL_w(PTS) = 155 \text{ dB re } 1 \mu Pa^2 s$$

$$SEL_w(TTS) = 140 \text{ dB re } 1 \mu Pa^2 s$$

The noise limits for behavioral effects are based on the weighted Root Mean Square Sound Pressure Level with an averaging time of 125 ms (Tougaard et al., 2014, 2015, 2016):

$$SPL_{w,rms} = 103 \text{ dB re } 1 \mu Pa$$

1.2 Sound Source Sercel Micro-GI Air Gun

The source wavelet is shown in the upper panel of Figure 1. In the second the source spectrum is given. This is contrasted with the hearing threshold of the VHF in the third panel. The signal is characterized by a frequency content of 1.3×10^2 Hz to 4.8×10^2 Hz at -6 dB relative to the maximum power, a pulse length of 4.0 ms and the following UNweighted source intensities at a reference distance of 1m:

$$SPL_{uw,0-peak}(1\text{ m}) = 226.0\text{ dB re } 1\text{ }\mu\text{Pa}$$

$$SEL_{uw}(1\text{ m}) = 196.0\text{ dB re } 1\text{ }\mu\text{Pa}^2\text{ s}$$

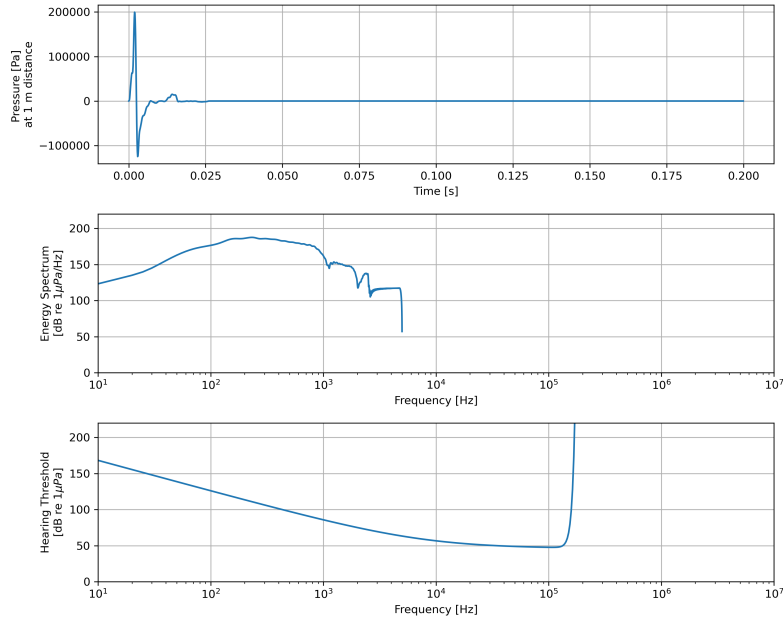
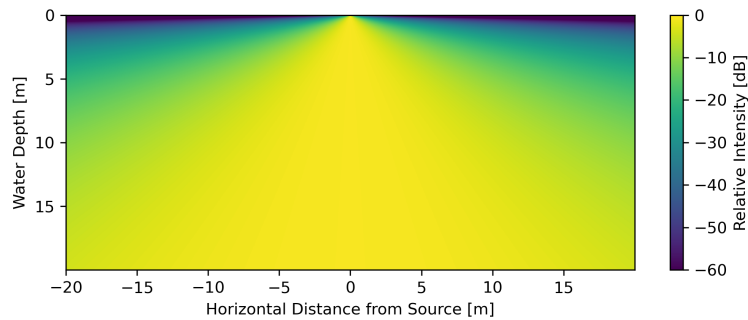


Figure 1: Source characteristics of the Sercel Micro-GI Air Gun; First panel: Time series of the wavelet record; Second panel: Source power spectrum; Third panel: Hearing threshold of the VHF.

For a single seismic source, the directivity is estimated with the Lloyds Mirror effect based on Carey (2009). The most relevant factors are the dominant frequency of 440.0 Hz and the source tow depth of 0.7 m. The resulting and applied directivity are shown in Figure 2.



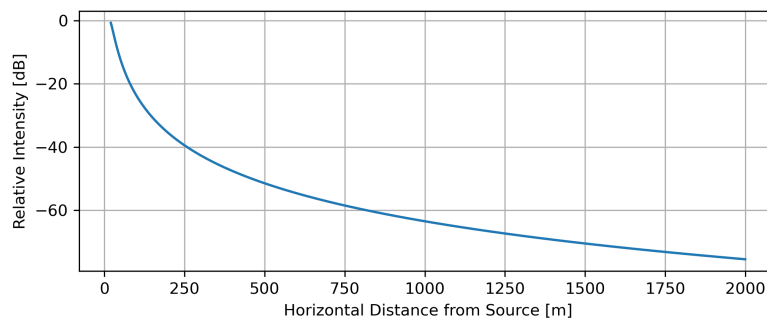


Figure 2: Upper panel: Calculated directivity, Lower panel: Directivity function applied to the exposure metric calculation.

2 Calculation Basics

For the calculation of the metrics, knowledge about the geometric spreading as shown in Figure 2, absorption and the filter function for the functional hearing group are relevant.

2.1 Geometric Spreading

To account for the geometric spreading, the ShallowWater-model is used (see e.g. Elmer et al., 2007; Bradley and Stern, 2008; Duncan and Parsons, 2011). For this geometric spreading model, the waterdepth 20 m and the safety threshold -6.6 dB are relevant.

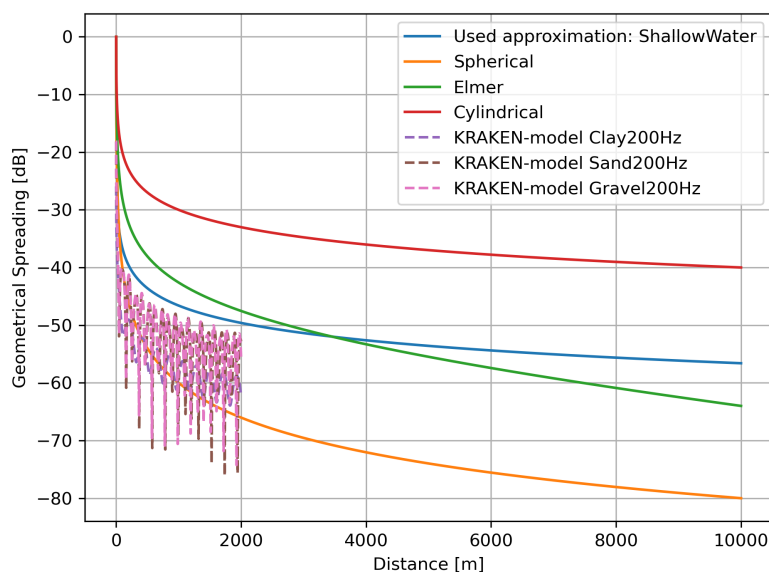


Figure 3: Geometric spreading approximations; The used approximation is drawn as a blue solid line.

2.2 Absorption

The absorption is calculated according to Ainslie and McCollm (1998) for the Baltic Sea and shown in Figure 3.

2.3 Filter Function

The frequency dependent filter function shown in Figure 4 is calculated from the weight function as specified in Southall et al. (2019).

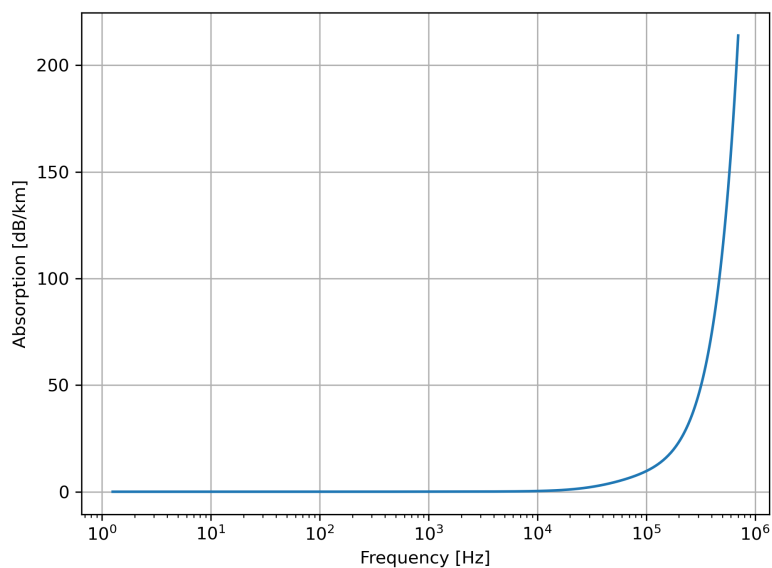


Figure 4: Absorption model.

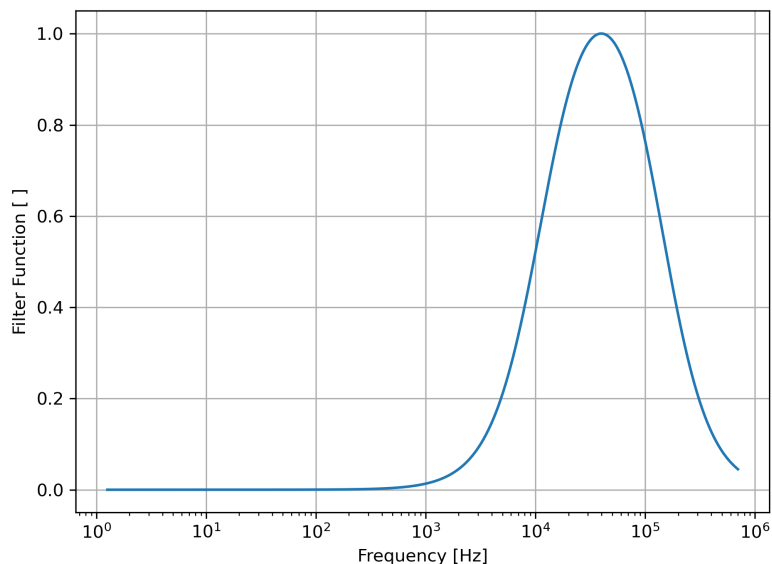


Figure 5: Frequency dependent filter function for the functional hearing group VHF.

3 Safety distances

The dual exposure metric unweighed SPL and weighted SEL are calculated to determine the safety distances. To give conservative distance estimates, the larger distance of the dual metric should be considered for both PTS and TTS.

3.1 SPL limits PTS and TTS

Based on the source signal, absorption and geometrical spreading, the PTS onset distance limit is 13.0 m, while the TTS onset distance limit is 20.0 m, as shown in Figure 5.

3.2 SEL limits PTS and TTS

Based on the source signal, absorption and geometrical spreading, the PTS onset distance limit is 0.0 m, while the TTS onset distance limit is 0.0 m. A stationary animal and varying passing distances, which are given on the x- axis of Figure 6, are considered. For the calculation of the cumulative SEL , a profile duration of 0.1511 h with a survey speed of 5.0kn and a shot repetition rate of 3.0s are considered.

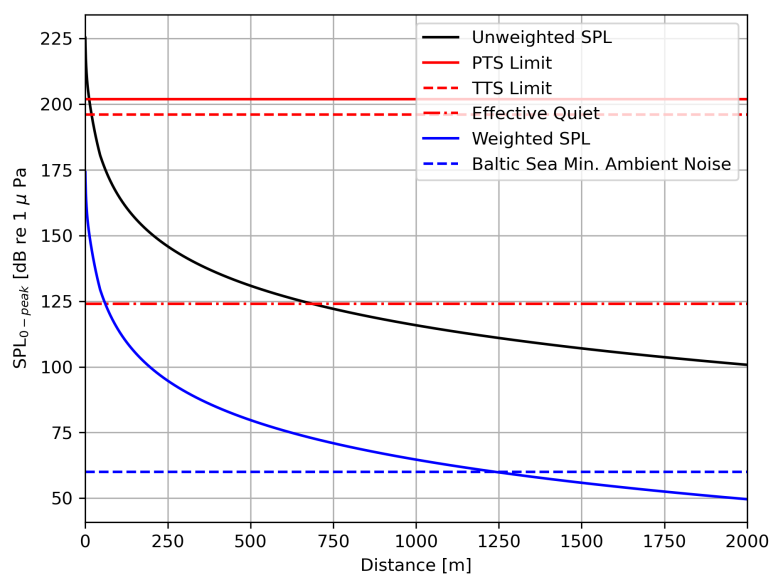


Figure 6: Reduction of the unweighted zero-peak SPL as a function of distance to the source with an overlay of the noise limits.

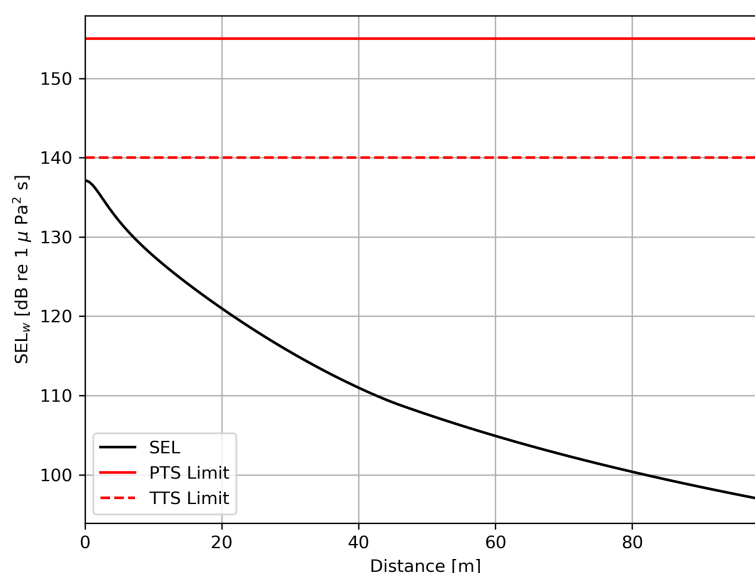


Figure 7: Reduction of the weighted and cumulative SEL as a function of the distance to a profile line assuming a stationary animal.

3.3 RMS-SPL limits Behavioural Impacts

Based on the source signal, absorption and geometrical spreading, the onset distance limit for behavioural effects is 22.0 m as shown in Figure 6.

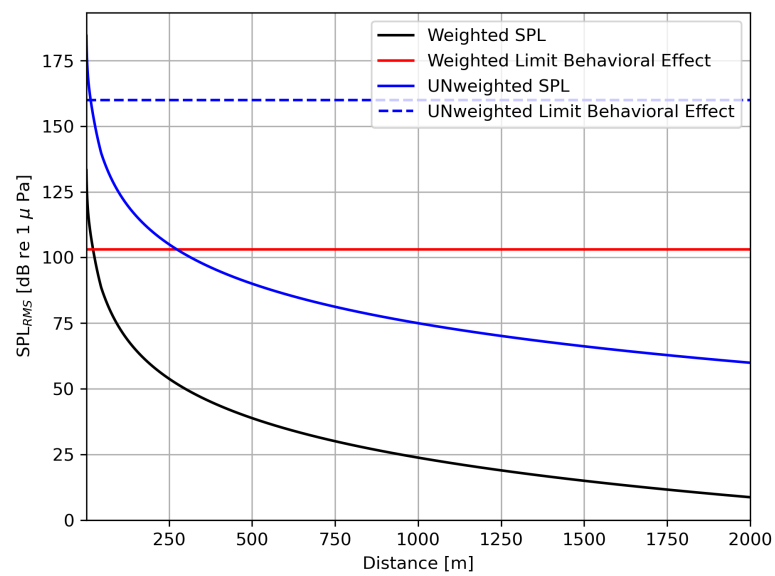


Figure 8: Reduction of the weighted RMS SPL with an averaging period of 125 ms as a function of distance to the source with an overlay of the noise limit.